

VL 211: Analog Circuit Design

Course Code: VL 211

Course Name: Analog Circuit Design

L-T-P-C: 3-0-2-4

Prerequisites: Network theory (EE 203), Applied Electronics (IC 161), Devices Electronics for Integrated Circuits (EE 311)

Intended for: BTech (EE)

Distribution: Core for B. Tech. (EE) & B.Tech (VLSI) and Elective for other UG Programs

Semester: Even/Odd

Approval: 61st BoA; 64th BoA (EE 211 Analog Circuit Design discontinued and replaced by VL 211)

Preamble and Objectives

Analog Circuit Design course is designed to build a strong foundation in both design and analysis of electronic circuits by mixture of lectures and having relevant experiments. This course syllabus offers conceptual understanding and mastery of the material by using modern examples to motivate and prepare IIT Mandi's undergraduate students for advanced courses and their careers. The course has unique problem-solving framework enables the students to deconstruct complex active circuits into small signal equivalent model that they are familiar with which builds the confidence and intuitive skills needed for success. To succeed in the practice of electronics in industry, students must develop the ability to think intuitively about circuits. They need to move beyond simply plugging and chugging numbers in equations and be prepared to face real design trade-offs. The course components also help students to develop intuitive techniques so they can design and implement circuits not just analyze them. Using many real-world examples and applications will motivate the student to understand the importance of today's microelectronics. The course is intended for advance stream of studies such as communication engineering, power electronics and very large scale integration (VLSI).

Lecture modules

- **PN junction, BJT, MOSFET and PN junction circuit and Network theory review:** [5 hrs]
- **BJT/MOS single stage amplifiers, cascade and cascodes:** Large signal and small-signal model, biasing, input and output impedance, operating point calculations and design, single ended BJT/CMOS amplifiers, cascade and cascode amplifiers [8 hrs]
- **BJT/MOS Current mirrors:** PVT independent normal current mirror, cascode current mirror, regulated current mirror and Wilson current mirror [3 hrs]
- **Differential Amplifiers:** MOS/BJT Differential Pair, qualitative large/small signal analyses, differential pairs with active loads and common-mode rejection [6 hrs]
- **BJT/MOS Frequency Response:** High frequency model of single ended and differential amplifiers, Frequency response: magnitude and phase plot calculations [6 hrs]
- **Feedback theory:** Properties of negative and positive feedback, loop gain calculations, types of amplifiers, voltage controlled current source (VCCS), current controlled current source (CCCS), current controlled voltage source (CCVS), voltage current voltage source (VCVS), stability analyses under negative feedback topology. [6 hrs]

- **Oscillators:** Barkhausen criteria of oscillation, bistable, monostable and astable multi-vibrators, LC, relaxation, phase shift and Colpitt oscillators [4 hrs]

Experiment modules

1. Introduction to laboratory with DSO and its advance features, XY-mode (Lissajous pattern), LTspice and required softwares,
2. Voltage amplifier circuit design and SPICE simulations
3. Understanding of I-V characteristics of BJT/MOS transistor using net-listing in SPICE simulations.
4. Common emitter amplifier design.
5. Oscillator: design of ring, LC and phase shift oscillators.
6. Project on analog circuit application

Text Books

1. *Fundamentals of microelectronics* by Behzad Razavi, Wiley, Apr. 2013.

Reference Books

1. *Microelectronic Circuits-Theory & Applications* by A.S. Sedra and K.C. Smith, 7th Edition, Oxford University Press, Jun. 2017.